INCH-POUND

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23 January 1989
SUPERSEDING
MIL-H-20194D
2 December 1982

MILITARY SPECIFICATION

HELIOTROPE, SURVEYING, BOX-TYPE, WITH CASE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a box-type heliotrope with carrying case.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-S-3534 - Surveying Instruments and Accessories, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN; NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

ME

TA13217E4800	- Heliotrope, Surveying, Box-Type with Case.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-JDT, Fort Belvoir, VA 22060-5606.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The heliotrope and case shall be in accordance with TA13217E4800, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operaton of the heliotrope. Any data (e.g., shop drawings,

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layouts, flow sheets, processing procedures, materials and material finishes, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 4.1.2).

3.3.1 Material deterioration prevention and control. The heliotrope shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the heliotrope may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the heliotrope may be newly fabricated from recovered materials to the maximum extent practicable, provided the heliotrope produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the heliotrope.

3.4 Dimensions. Tolerance and gages shall conform to the limits specified herein and on the applicable drawings.

3.5 Operational performance.

3.5.1 Mechanical operating conditions. All operating mechanical features shall be movable without binding of parts within the temperature extremes specified in 3.6.1.

3.5.2 Alignment. The front and rear sights shall be mounted above the alignment rings so that a line passing through the centers of the sights shall be parallel within 0.3969 mm to a line passing through the centers of the alignment rings.

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3.5.3 Mirrors. Mirrors shall be second-surface silvered on glass. The reflecting surface shall be dense, chemically deposited silver, protected by not less than a 0.0254 mm-thick electrodeposited copper coating. The back and edge of each mirror shall be coated with a baked-on, weather-resistant lacquer.

3.6 Environmental conditions.

3.6.1 Operating temperature. The heliotrope shall perform as specified herein in any ambient temperature from +43 °C to -32 °C.

3.6.2 Storage temperature. The heliotrope shall not be damaged by storage in any ambient temperature from +63 °C to -33 °C.

3.7 Identification marking. The heliotrope and case shall be identified in accordance with MIL-STD-130 and as specified herein.

3.8 Workmanship. All parts, components, and assemblies of the heliotrope and case shall be clean and free from corrosion, cracks, scratches, or other defects or extraneous material that will affect appearance or serviceability. All edges shall be rounded or beveled. There shall be no evidence of damaged parts or assemblies that would make the item unsatisfactory for its intended purpose.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The first article shall be tested as specified in 4.5.2. Failure of any tests shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. Acceptable quality level (AQL) shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 1.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The heliotropes and cases shall be examined as specified herein for the following defects.

Major

101. Material not as specified (see 3.3).
102. Material not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).
103. Dissimilar metals, as defined in MIL-STD-889, are not effectively insulated from each other (see 3.3.1.1).
104. Contractor does not have documentation available for identification of material, material finishes, or treatment (see 3.3.1.2).
105. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the heliotrope (see 3.3.2).
106. Dimensions not as specified (see 3.4).
107. Mirrors not as specified (see 3.5.3).

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Minor

201. Identification marking missing, incomplete or illegible (see 3.7).
202. Workmanship not as specified (see 3.8).

4.5.2 Tests.

4.5.2.1 High temperature. The heliotrope shall be tested as specified in MIL-STD-810, procedure II, method 501.1. The maximum storage temperature shall be 63 °C and the maximum ambient operating temperature shall be 43 °C. Operate the heliotrope as specified in 4.5.2.3. Nonconformance to 3.5.1 or 3.5.2 shall constitute failure of this test.

4.5.2.2 Low temperature. The heliotrope shall be tested as specified in MIL-STD-810, procedure I, method 502.1. The minimum storage temperature shall be -33 °C and the minimum ambient operating temperature shall be -32 °C. Operate the heliotrope as specified in 4.5.2.3. Nonconformance to 3.5.1 or 3.5.2 shall constitute failure of this test.

4.5.2.3 Alignment. The sights of the front and rear vanes shall be aligned by reflecting a light beam through the principal mirror to a nearby surface perpendicular to the line of sight. The shadow of the rear alignment ring shall fall on the annular band of the front ring. Evidence of the line of sight and the beam of light not parallel to within 0.3969 mm shall constitute failure of this test.

4.6 Inspection of packaging. Preservation, packing, and marking for level A, B, or C shall be examined and tested in accordance with the quality assurance provisions of MIL-S-3534.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each heliotrope in its case shall be preserved in accordance with the level A preservation requirements of MIL-S-3534.

5.1.2 Level B. Each heliotrope in its case shall be preserved in accordance with the level B preservation requirements of MIL-S-3534.

5.1.3 Level C. Each heliotrope in its case shall be preserved in accordance with the level C preservation requirements of MIL-S-3534.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Heliotropes, preserved as specified in 5.1, shall be packed together in accordance with the level A packing requirements of MIL-S-3534.

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5.2.2 Level B. Heliotopes, preserved as specified in 5.1, shall be packed together in accordance with the level B packing requirements of MIL-S-3534, except boxes conforming to PPP-B-636, V3c, style optional, may also be used. Box closure and sealing shall be in accordance with the appendix to the box specificaiton.

5.2.3 Level C. Heliotropes, preserved as specified in 5.1, shall be packed together in accordance with the level C packing requirments of MIL-S-3534.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The heliotrope is intended for use as a target in daylight triangulation and reconnaissance surveys.

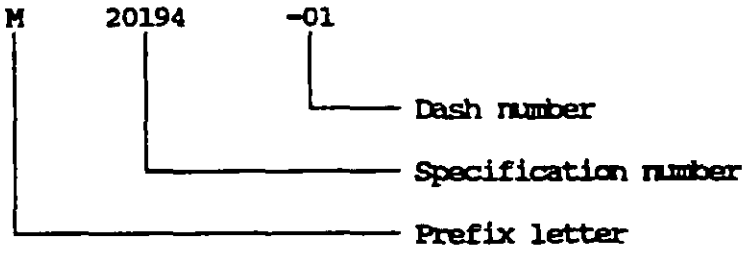
6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. Part number (see 6.4).
d. Timeframe required for submission of first article and number of units required (see 3.2).
e. Level of preservation and packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.4 Part or identifying number (PIN). The military specification part number for the Heliotrope, Surveying, Box-Type, with case should be as follows (see 6.2).



6.5 Subject term (key word) listing.

Box-type heliotrope
Surveying box-type heliotrope
Surveying heliotrope
Target heliotrope

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
AF - 99

Review activities:
AF - 82

Preparing activity:
Army - ME

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